1. What is Object-Oriented Programming?

Follow-up Questions:

- Why do we use OOP instead of procedural programming?
- Can you give a real-life example of OOP?

Expected Topics:

- Objects and Classes
- Code Reusability
- Modularity
- Maintainability

2. What is the difference between a Class and an Object?

Follow-up Questions:

- Can an object exist without a class?
- How many objects can be created from one class?
- Where are objects stored in memory?

Expected Topics:

- Blueprint vs Instance
- Memory Allocation
- Constructors

3. Explain the four pillars of Object-Oriented Programming.

Follow-up Questions:

- Which pillar provides security?
- Which pillar helps in code reuse?
- Give one real-life example of each.

Expected Topics:

- Encapsulation
- Inheritance
- Polymorphism

- Abstraction

4. What is Encapsulation?

Follow-up Questions:

- How is encapsulation achieved?
- Why are variables declared private?
- What are getters and setters?
- Can encapsulation exist without inheritance?

Expected Topics:

- Data Hiding
- Access Modifiers
- Getter/Setter Methods

5. What is Abstraction?

Follow-up Questions:

- Difference between abstraction and encapsulation.
- How do abstract classes work?
- Can an abstract class have constructors?
- Can it contain implemented methods?

Expected Topics:

- Hiding Implementation
- Abstract Class
- Interface

6. What is Inheritance?

Follow-up Questions:

- Types of inheritance.
- Advantages and disadvantages.
- Why doesn't Java support multiple inheritance using classes?
- Difference between "extends" and "implements".

Expected Topics:

- IS-A Relationship
- Code Reuse
- Parent and Child Classes

7. What is Polymorphism?

Follow-up Questions:

- Compile-time vs Runtime polymorphism.
- Method overloading vs overriding.
- Can constructors be overloaded?
- Can static methods be overridden?

Expected Topics:

- Overloading
- Overriding
- Dynamic Method Dispatch

8. Explain Method Overloading and Method Overriding.

Follow-up Questions:

- Which one is compile-time?
- Which one is runtime?
- Can return type alone differentiate overloaded methods?
- Can private methods be overridden?

Expected Topics:

- Same Method Name
- Parameters
- Inheritance
- Runtime Binding

9. What are Constructors?

Follow-up Questions:

- Types of constructors.
- Constructor overloading.
- Difference between constructor and method.
- Can constructors be inherited?
- Can constructors be private?

Expected Topics:

- Default Constructor
- Parameterized Constructor
- Object Initialization

10. What is the difference between an Interface and an Abstract Class?

Follow-up Questions:

- When should you choose an interface?
- Can an interface have constructors?
- Can an abstract class have instance variables?
- Can a class implement multiple interfaces?

Expected Topics:

- Multiple Inheritance
- Default Methods
- Abstract Methods
- Design Decisions

Campus Credentials

Bonus Rapid-Fire Questions

1. What is the this keyword?
2. What is the super keyword?
3. Can we create an object of an abstract class?
4. Can an interface have variables?
5. What is constructor chaining?
6. What is dynamic binding?
7. What is static binding?
8. What is object cloning?
9. What is object composition?
10. What is the difference between HAS-A and IS-A relationships?